

IIT Bhilai Senior Survey 2025–26: A Data-Driven Analysis of Student Life and Outcomes

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Abstract

This paper presents the IIT Bhilai Senior Survey 2025–26, a data-driven study of the experiences, habits, and future plans of the graduating batch at IIT Bhilai. Inspired by the annual Insight Senior Survey conducted at IIT Bombay, this survey was adapted to reflect the unique context of a newer IIT still developing its identity and infrastructure. A total of 97 responses were collected across 11 departments covering academics, career aspirations, lifestyle, and campus life. Key findings include that 61% of respondents are heading directly to jobs, salary and work culture are rated as the most important job factors, and campus infrastructure received an average rating of 3.0 out of 5.

A logistic regression model was trained to predict whether a graduating student would pursue employment or higher studies, achieving 72.2% accuracy. The interactive dashboard for this survey is available at: <https://shubham7094.github.io/IITBH-Senior-Survey/>

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1 Introduction

Graduating students at engineering institutions accumulate a wide range of academic, social, and personal experiences over four years. However, these experiences are rarely systematically documented or analyzed. Understanding such patterns can provide meaningful insights for students, faculty, and institutional development.

This project was inspired by the Insight Senior Survey conducted at IIT Bombay. Since no equivalent survey existed for IIT Bhilai, this study aimed to design and conduct a similar survey tailored to the context of a newer IIT still evolving in terms of infrastructure, reputation, and academic offerings.

The primary research questions guiding this study were:

- What are the academic and lifestyle patterns of IIT Bhilai's graduating batch?
- What factors influence post-graduation career decisions?
- How do students evaluate their overall experience at the institute?

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2 Data Collection

The survey was conducted using a Google Form consisting of 55 questions across five categories: demographics, academics, career plans, lifestyle, and campus experience.

The form was distributed through batch communication channels including WhatsApp groups and personal networks during April 2026. A total of 97 valid responses were collected across 11 departments.

The largest representation came from Electrical Engineering and Computer Science Engineering (20 each), followed by Data Science and AI and Mechanical Engineering (16 each). The sample consisted of 82 male and 15 female respondents, broadly reflecting the institute's demographic composition.

3 Data Cleaning and Preparation

The raw dataset required several preprocessing steps before analysis.

Categorical inconsistencies were addressed by standardizing similar responses into unified labels. For example, multiple variations of non-vegetarian dietary responses were merged into a single category. Similarly, overlapping entries in grade-related fields were normalized.

Missing values in Likert-scale rating columns were imputed using column-wise median values, which is appropriate for ordinal data. Column names were also simplified for ease of analysis.

Importantly, no rows were removed during preprocessing, preserving all 97 responses.

4 Results and Analysis

4.1 Academic Patterns

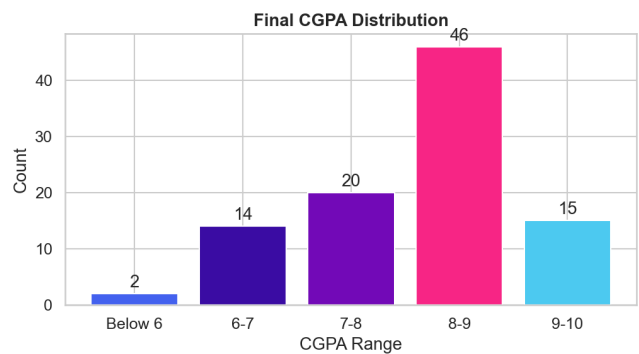


Figure 1: Final CGPA distribution across 97 respondents.

The CGPA distribution is heavily concentrated in the 8–9 range, accounting for approximately 47% of respondents. Very few students fall below a CGPA of 6, indicating an overall high-performing cohort.

However, study habits reveal a contrasting trend: 69% of students reported studying primarily before exams rather than maintaining a consistent study schedule. This suggests reliance on short-term preparation strategies.

Additionally, more than half of the respondents reported witnessing academic honor code violations without reporting them, indicating a gap between institutional policy and student behavior.

4.2 Career Preferences

Immediate Plans After Graduation

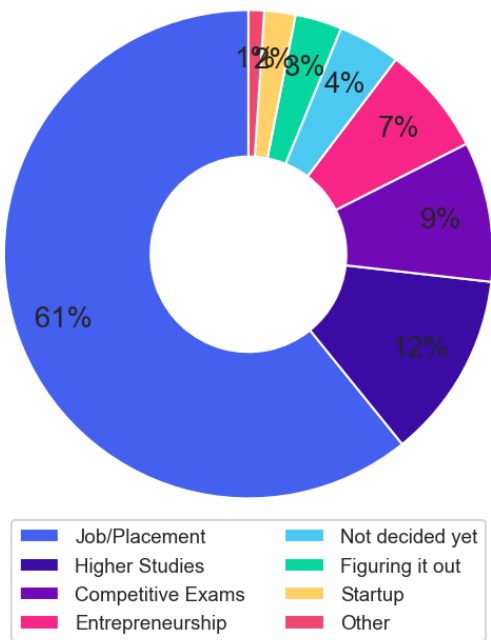


Figure 2: Immediate post-graduation plans of respondents.

A majority (61%) of students intend to pursue job placements immediately after graduation, while only 12% opt for higher studies. This trend is somewhat lower in higher-study preference compared to older IITs, suggesting a stronger inclination toward immediate employment.

Among job-related factors, salary (4.11/5) and work culture (4.02/5) were rated highest, followed by interest alignment. This indicates a pragmatic approach toward career decisions.

4.3 Career Alignment

Interest in core engineering careers is mixed, with a notable proportion of students expressing neutrality or disinterest. This aligns with broader trends where students transition toward non-core roles.

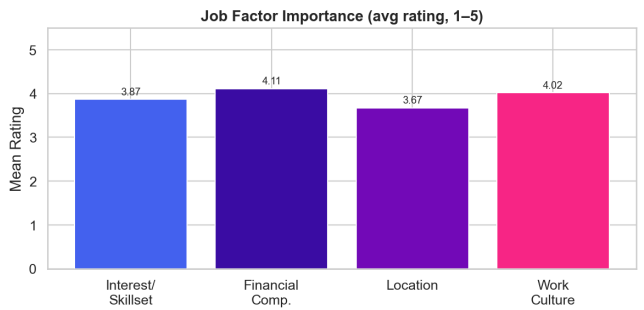


Figure 3: Average importance ratings of job factors (1–5 scale).

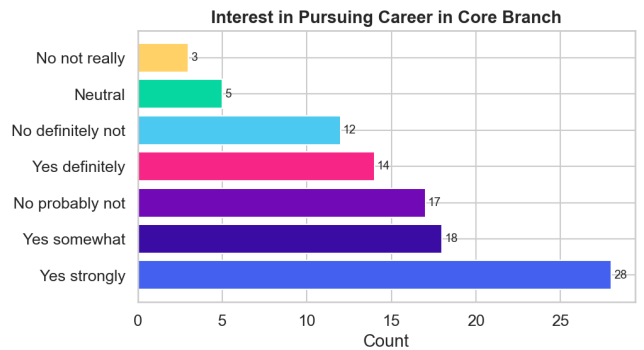


Figure 4: Interest in pursuing a career in core branch.

4.4 CGPA vs Career Choices

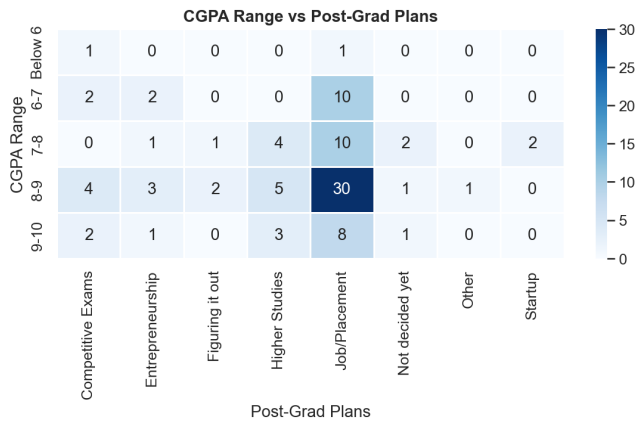


Figure 5: Relationship between CGPA range and post-graduation plans.

Students with higher CGPA show a slightly greater inclination toward higher studies, although job preference remains dominant across all CGPA ranges.

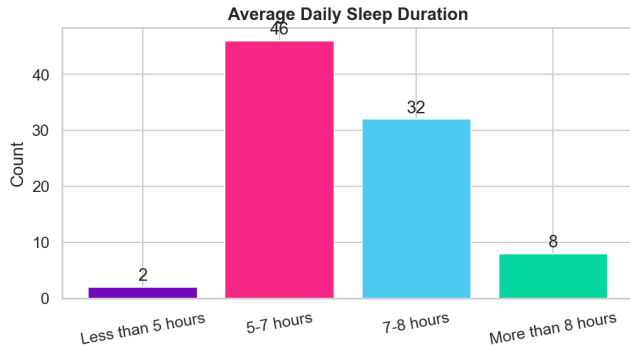


Figure 6: Average daily sleep duration among respondents.

4.5 Lifestyle Patterns

Approximately 80% of respondents report sleeping between 5–8 hours, indicating moderate sleep patterns.

Substance usage remains limited, with 55% of students reporting never having consumed alcohol and 75% reporting never having smoked.

4.6 Satisfaction Metrics

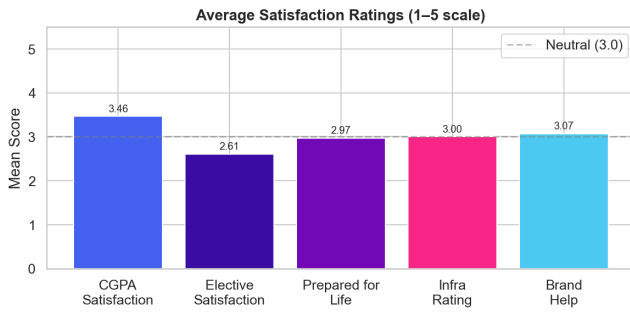


Figure 7: Average satisfaction ratings across key dimensions (1–5 scale).

Elective satisfaction (2.61/5) is the lowest-rated dimension, suggesting dissatisfaction with course variety or quality. Infrastructure received a neutral average rating of 3.0/5, indicating mixed perceptions among students.

4.7 Demographics

Respondents come from diverse regional backgrounds, with Tier-2 cities being the most common origin.

5 Dashboard

An interactive dashboard was developed using Chart.js and deployed via GitHub Pages. The dashboard integrates all cleaned data into a single HTML file and provides interactive visualizations across multiple sections.

The dashboard can be accessed at: <https://shubham7094.github.io/IITBH-Senior-Survey/>

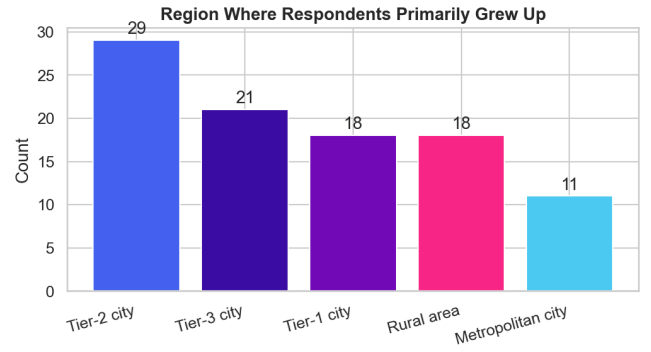


Figure 8: Distribution of respondents by region of upbringing.

6 Methodology and Model Building

6.1 Problem Formulation

The goal of the model was to predict whether a student would pursue a job or higher studies. This was formulated as a binary classification problem.

$$y = \begin{cases} 1, & \text{Job/Placement} \\ 0, & \text{Higher Studies} \end{cases}$$

The dataset was filtered to include only these two categories, resulting in 71 samples.

6.2 Model

Logistic regression was used due to its interpretability and suitability for binary classification.

The probability of choosing a job is modeled as:

$$P(y = 1|x) = \frac{1}{1 + e^{-(\beta_0 + \beta^T x)}}$$

where x is the feature vector and β represents model coefficients.

The loss function minimized is:

$$L = - \sum_{i=1}^n [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)] + \lambda ||\beta||^2$$

The second term represents L2 regularization to prevent overfitting.

6.3 Results

- Accuracy: 72.2%
- ROC-AUC: 0.62
- Cross-validation accuracy: 77.4% $\pm$ 5.5%

6.4 Interpretation

Key influencing factors include:

- Lower importance given to classes correlates with higher likelihood of pursuing higher studies
- Strong alignment with interest increases likelihood of choosing jobs
- Work culture preference significantly impacts decisions

6.5 Limitations

- Small sample size
- Class imbalance
- Self-reported data bias

7 Conclusion

This study provides a structured overview of the IIT Bhilai graduating batch. The results highlight a cohort that is career-focused, academically capable, and socially connected.

While job preference dominates, areas such as elective quality and infrastructure require improvement. The study also demonstrates the usefulness of data-driven institutional feedback.

Future work can expand dataset size and incorporate more advanced modeling techniques.

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